



Hands-on Principal Architect · 19+ yrs · AI-Native + Agentic Systems · Vector Retrieval + Semantic Search · Python (FastAPI, LangChain, LangGraph) · Neptune + GNN · Java + Spring Boot · HIPAA, ISO 27001 · AI-Leveraged Delivery (Claude Code + OpenAI Codex)

SUMMARY

Hands-on principal architect, 19+ years, building artificial intelligence (AI)-native platforms where the retrieval path is the product. Most recently ten analytics and AI services for a chronic-care platform on Amazon Web Services (AWS): Elasticsearch dense_vector retrieval with self-hosted Amazon SageMaker embedders, a Python orchestration layer on FastAPI, LangChain and LangGraph, and deterministic code owning every number, all reading 9–10 years of patient history. 15 shared platform components now carry the isolation, statistics and correction contracts, so a sibling query inherits a fix rather than carrying its own copy, and protected health information never leaves the client boundary — embeddings and Llama 3.1 self-hosted for every model call. Underneath sits 11 years of Java and Spring Boot across three eras and multi-tenant software as a service (SaaS) consolidation across approximately 300 enterprise instances. Kubernetes and Terraform adopted as a company-wide standard, infrastructure cost down 25–30%, deploys under 10 minutes, on a 20-engineer team running Claude Code and OpenAI Codex under review.

Key Skills

Technology Leadership & Strategy

- Technology Strategy & Architecture
- AI-Native & Agentic Platform Strategy
- Horizontal Architecture Roadmap & Cross-Organisation Standards
- Decision-Intelligence Platforms (Event-Sourced + Graph + AI)
- Build-versus-Buy Advisory (AI and Machine Learning Platform Investment)
- Engineering Team Building & Organisational Scaling
- Cross-Functional Leadership & Stakeholder Alignment

Engineering Excellence

- Vector Retrieval & Semantic Search (Elasticsearch dense_vector, Filtered Nearest-Neighbour, Self-Hosted Embeddings)
- Retrieval Data Access Layers (Cache-Aside Serving, Filtering, Sorting, Custom Relevance Ranking)
- Agentic & Large Language Model (LLM) Service Interfaces (LangChain, LangGraph, Model Context Protocol (MCP) Servers, Pydantic v2 Structured Output)
- Python Services in Production (FastAPI, Django, Pydantic v2)
- Query Routing Across Reusable Archetypes (Retrieval, Cohort Aggregation, Sequence Mining, Graph Traversal)
- Self-Hosted Models Inside the Virtual Private Cloud (VPC) (Ollama, Llama 3.1, SageMaker Embeddings, Self-Hosted LangSmith)
- LLM Productionisation (Guardrails, Field-Level Claim Grammar, Replay Determinism)
- Shared Platform Standards & Application Programming Interfaces (APIs) (15 Reusable Components, Versioned Contracts, Backward Compatibility)
- AI / Machine Learning in Product (Graph Neural Network — GNN — on Amazon Neptune + Gremlin, Clinical LLM Workflows)
- Vector & Columnar Data Stores (Elasticsearch dense_vector, DuckDB + Parquet, Redis, Amazon Neptune)
- Java + Spring Boot Microservices (Java 8/17, Spring Boot 2.x, JSP — 11 years across three eras)
- Multi-Tenant SaaS Architecture (Shared-Schema, Per-Tenant Isolation, Regulated Carve-Outs)
- Event Sourcing + Command Query Responsibility Segregation (CQRS) + Snapshot Exporters
- Leakage-Safe Machine Learning (As-Of Featurisation, Point-in-Time Correctness, Outcome-Maturity Windows)
- Cloud Infrastructure & Platform Engineering (AWS — Lambda, SageMaker, Aurora, DynamoDB, Kubernetes, Terraform)
- Cross-Team Technical Leadership (6 sub-teams, company-wide standards)
- Hands-on Player-Coach Delivery (AI-Leveraged via Claude Code + OpenAI Codex)

Business & Delivery

- Design-Packet & Pre-Registered Acceptance-Criteria Discipline (Structured Adversarial Design Review)
- Security, Compliance & Data Privacy (Health Insurance Portability and Accountability Act (HIPAA) Pipelines in an ISO 27001-Certified Environment, Least-Privilege)
- Performance Optimization & Cost Management
- Enterprise Onboarding & Consolidation at Multi-Hundred-Instance Scale
- Distributed United States–India Delivery Teams
- Accessibility Compliance (Web Content Accessibility Guidelines (WCAG) 2.2 Level AA)

PROFESSIONAL EXPERIENCE

Principal Software Architect	Mar '25 - Apr '26
Voltuswave Technologies India Pvt Ltd	Hyderabad
• Engineered a retrieval-augmented generation (RAG) pipeline on Elasticsearch dense_vector filtered nearest-neighbour search across 80+ chronic conditions, keeping Amazon Neptune off the serving path	

- Tuned precedent retrieval on **Elasticsearch dense-vector** search with self-hosted **Amazon SageMaker** embedders, designed against warm-path **P99 under 200 ms** and **0.75 normalised discounted cumulative gain**
- Assembled a **10-query** decision-intelligence platform in **Python** — **FastAPI, LangChain, LangGraph** — on **4 reusable archetypes**: Retrieval, Cohort Aggregation, Sequence Mining, Graph Traversal
- Secured protected health information inside the client **virtual private cloud** by self-hosting embeddings and **Llama 3.1** for every model call
- Pioneered a Context Graph on **Amazon Neptune** with **GNN** models, powering virtual-care decision intelligence for **10,000+ patients** on a chronic-care platform
- Promoted **15 shared platform components** out of per-query code — isolation, statistics, correction contracts — **14 landed**, every query now binding to the shared version
- Brokered company-wide adoption of **Go, DynamoDB, Kubernetes** and **Terraform** as platform standards, **Go and DynamoDB across 5 sub-teams**, backed by design-doc discipline

Lead SDE/Sr. Staff Engineer / Staff Software Engineer

Dec '19 - Feb '25

Deque Software

Hyderabad

- Implemented multi-tenant isolation across **approximately 300 enterprise instances**, adding an organisation ID to every table and resolving tenant from the **JSON Web Token (JWT)**
- Exposed shared platform APIs to **20+ enterprise clients** — Jira integration, API versioning, customer-specific frontends on one **Spring Boot** backend, backward compatibility
- Steered **70–80%** of customers onto shared **axe Monitor** instances on **Spring Boot 2.x and Java 8/17**, one codebase also serving regulated banks on dedicated databases
- Forged central authentication in **Java and Spring Boot**, migrating every instance from **Keycloak** to **Red Hat single sign-on** — users, roles, clients, groups, zero disruption
- Championed decomposition of a monolith into **10–15 services**, splitting on independent scaling and database ownership rather than on team boundaries
- Productized a billing and subscription service consumed by **every product** in the portfolio, porting it from **Node.js/TypeScript** to **Java and Spring Boot**
- Stewarded a **zero-downtime** upgrade from **PostgreSQL 9.6** to **Amazon Aurora PostgreSQL 13.x** across every axe Auditor instance

Software Engineer III

Aug '18 - Jul '19

Rocket Software

Bangalore

- Anchored decomposition of **IBM Db2 Configuration Manager** into **Java and Spring Boot** microservices, defining boundaries across **z/OS and Linux, UNIX and Windows** platforms
- Coached a **5-engineer** team through delivery planning, code review and performance appraisals while remaining hands-on as an individual contributor
- Surfaced defect hot-spots through **REST** interfaces and service isolation, reducing reported bugs **over 50%**
- Partnered with product to extend platform support to **Windows** and Linux — Ubuntu, CentOS and **Red Hat Enterprise Linux (RHEL)** — raising sales **over 20%**

Co-Founder and VP of Technology

Mar '17 - Apr '18

Voltuswave Technologies India Pvt Ltd

Hyderabad

- Co-founded VoltusWave and launched its first **application Platform-as-a-Service (aPaaS)** product for web and Android, winning **8 early enterprise clients**
- Mobilized an engineering organisation from **0 to 35** in 13 months, mentoring freshers and senior engineers while establishing delivery process
- Helmed development, deployment and operations on **Amazon Elastic Container Service**, setting the architecture roadmap every subsequent product inherited

Senior Software Engineer

Mar '16 - Mar '17

Teletext India Private Limited, Hyderabad

Hyderabad

- Codified a scatter-gather integration across **25 hotel suppliers** on a **MuleSoft enterprise service bus**, succeeding where a prior **.NET 2.0** upgrade had failed
- Cut operational cost **46%** by retiring a third-party search product for an in-house **Elasticsearch** retrieval tier — cache-aside reads, **20-minute** time-to-live, filtering and sorting
- Reconciled supplier hotel identifiers onto **GIATA MultiCodes**, automating the pipeline on **AWS Lambda** with human approval retained, growing revenue **8% month over month**
- Lifted hotel booking conversions **65%** by widening matched inventory and ranking contracted properties first across **2–3 million daily requests**

Technical Architect

May '15 - Nov '15

CURA Software Solutions

Hyderabad

- Introduced **Linux/Unix deployment** for the Governance, Risk & Compliance platform, cutting running costs and winning **12 new customers within 2 months**
- Ported the data model from **Microsoft SQL Server** to **MySQL**, enabling the low-cost Linux deployment

Technical Lead

Jul '13 - May '15

innRoad India Hotel Software Pvt. Ltd.

Hyderabad

- Converted a **legacy ASP.NET 2.0** application into a modern **single-page application** using Knockout.js and a Boilerplate JavaScript framework
- Productized **REST-based WCF** services that enabled and monetised customer integrations
- Codified engineering standards and mentored the team through a full **product-modernisation effort**

Software Architect

Jun '12 - May '13

McDonald's Corporation

OakBrook, IL

- Modeled the schema and governance framework for an enterprise Master Data Management (MDM) program
- Forged interactive user-interface screens with drill-down, dynamic drop-downs, and full row-level editing
- Partnered with data owners to deliver a robust enterprise data-governance system across the organisation

Senior Java Developer	Apr '08 - May '12
El Paso Corporation	Houston, TX
- Rebuilt a legacy Citrix-hosted Java application into a modern .NET 2.0 stack on SQL Server 2005 - Steered a scalable, maintainable .NET platform that replaced the outdated system while preserving core business functions - Anchored large-scale enterprise modernisation for a high-stakes energy-sector client	
.NET Developer	Apr '07 - Apr '08
LyntonWeb	Houston, TX
Developed Paymaster, a PayPal-style payments application, for a Mexican financial institution	

EDUCATION

PG Diploma in Artificial Intelligence and Machine Learning	Jan '21
IIIT Bangalore	
Master's in Computer Science	Aug '04 - Dec '07
University of Houston	Houston, Texas

CERTIFICATION

PG Diploma in Artificial Intelligence and Machine Learning, IIIT Bangalore